

ALVDALEN, Sweden

WMO: 023210

Lat: 61.2536N

Lon: 14.0355E

Elev: 830

StdP: 14.26

Time Zone: 1.00 (EUC)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-11.0	-6.5	-15.8	2.4	-10.5	-10.9	3.2	-6.1	15.4	36.4	13.3	33.5	1.7	20	0.611

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.8	79.0	62.2	75.7	61.4	72.5	59.9	65.6	74.7	63.5	71.6	61.5	69.0	6.1	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
62.2	86.4	68.2	60.3	80.5	66.8	58.4	75.2	64.8	30.8	74.5	29.3	71.7	27.8	69.1	72.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 13.5	11.5	9.7	DB	-17.1	82.7	4.5	2.5	-20.4	84.5	-23.0	86.0	-25.6	87.3	-28.9	89.1	(4)
(5)			WB	-17.5	67.7	4.1	2.2	-20.4	69.3	-22.8	70.6	-25.1	71.8	-28.1	73.4	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	38.2	17.2	20.2	27.9	37.9	47.6	54.8	60.0	56.4	48.6	38.1	28.9	20.1		
	DBStd	16.65	11.66	10.02	8.76	6.10	6.80	5.53	5.10	4.93	5.56	6.92	8.26	11.62		
	HDD50	5076	1016	835	685	365	130	16	1	7	89	371	633	928		
	HDD65	9785	1481	1255	1150	813	541	307	169	268	492	833	1083	1393		
	CDD50	786	0	0	0	2	54	161	312	207	47	3	0	0		
	CDD65	19	0	0	0	0	0	2	15	2	0	0	0	0		
	CDH74	459	0	0	0	0	26	97	286	50	0	0	0	0		
	CDH80	60	0	0	0	0	0	9	49	2	0	0	0	0		
	WSAvg	3.7	3.0	3.1	4.0	4.1	4.6	4.6	4.0	3.5	3.6	3.4	3.1	3.0		
	PrecAvg	26.0	1.6	1.1	1.0	1.6	2.0	2.9	3.7	3.3	2.6	2.4	2.1	1.7		
(15) Precipitation	PrecMax	36.1	3.5	2.0	2.1	4.6	4.4	6.1	7.0	6.8	5.8	5.8	5.7	2.7		
	PrecMin	20.8	0.4	0.2	0.2	0.3	0.2	0.3	0.9	1.4	0.6	0.7	0.8	0.3		
	PrecStd	3.6	0.9	0.4	0.5	1.1	1.0	1.2	1.4	1.3	1.2	1.3	1.1	0.7		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	42.0	44.1	54.3	66.8	77.5	81.2	84.6	79.4	70.4	59.7	50.7	45.7		
		MCWB	38.6	38.5	44.6	51.1	58.0	60.9	65.9	63.8	57.6	53.4	48.7	42.0		
	2%	DB	39.2	40.2	48.9	62.0	72.8	77.0	80.9	74.8	66.1	55.3	46.7	41.5		
		MCWB	36.4	36.3	40.9	48.4	56.1	60.2	64.3	62.8	56.3	50.4	44.4	39.2		
	5%	DB	36.1	36.7	44.5	57.0	68.3	73.3	77.1	71.4	62.6	52.3	42.9	38.4		
		MCWB	34.0	34.0	37.7	45.6	53.8	58.8	62.9	61.1	55.5	48.4	41.1	36.5		
	10%	DB	33.1	34.0	40.9	52.5	62.8	68.6	73.4	67.9	59.3	49.6	40.0	35.0		
		MCWB	31.7	32.0	35.4	42.7	51.8	56.4	61.6	59.4	53.4	46.7	38.6	33.8		
	0.4%	WB	39.4	39.4	44.6	52.8	60.1	64.8	68.2	66.9	60.5	54.8	48.9	42.8		
		MCDB	41.2	42.4	52.6	63.4	71.4	74.8	78.8	73.7	66.9	58.0	50.6	45.2		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	36.7	36.9	41.0	49.5	57.9	61.9	66.4	64.8	58.3	51.6	44.7	39.5		
		MCDB	38.6	39.7	48.0	60.6	69.6	72.0	76.8	71.6	63.8	54.4	46.2	40.9		
	5%	WB	34.3	34.4	38.3	46.2	55.3	59.9	64.6	62.6	56.2	49.2	41.5	36.6		
		MCDB	35.9	36.5	43.5	55.8	65.0	70.5	74.2	68.2	61.0	51.6	42.5	38.0		
	10%	WB	31.9	32.0	36.0	43.5	52.7	57.8	62.6	60.6	54.0	46.9	38.7	33.8		
		MCDB	33.1	33.6	40.4	50.8	61.9	67.3	70.7	66.3	58.3	49.2	39.8	34.8		
(35) Mean Daily Temperature Range	5% DB	MDBR	14.0	16.6	19.8	21.4	21.3	20.7	20.8	19.7	18.7	15.0	11.0	12.8		
		MCDBR	13.8	17.9	23.4	31.8	32.7	29.8	28.5	26.2	24.4	19.2	14.1	15.2		
		MCWBR	12.5	14.1	16.2	18.9	17.1	15.4	13.9	15.1	15.8	14.4	12.5	13.6		
	5% WB	MCDBR	13.2	15.8	21.8	29.4	26.3	25.6	24.9	21.5	21.7	16.5	13.1	14.8		
		MCWBR	12.2	12.9	15.4	18.3	14.2	13.8	13.3	13.6	14.9	13.2	12.0	13.5		
(40) Clear-Sky Solar Irradiance	taub		0.229	0.277	0.306	0.342	0.348	0.343	0.361	0.352	0.319	0.304	0.249	0.348		
	taud		2.162	2.279	2.317	2.372	2.444	2.485	2.447	2.464	2.541	2.485	2.243	2.775		
	Ebn at Noon		165	226	259	270	278	281	272	265	256	218	154	118		
	Edh at Noon		11	20	27	31	31	31	31	28	22	16	10	7		
(44) All-Sky Solar Radiation	RadAvg		88	300	799	1277	1491	1676	1525	1178	770	350	107	49		
	RadStd		13	56	80	134	151	129	194	111	101	60	28	6		

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (48 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page